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Generative Map Priors for Collaborative BEV Semantic Segmentation

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Abstract

Collaborative perception aims to address the constraint of single-agent perception by exchanging information among multiple agents. Previous works primarily focus on collaborative object detection, exploring compressed transmission and fusion prediction tailored to sparse object features. However, these strategies are not well-suited for dense features in collaborative BEV semantic segmentation. Therefore, we propose CoGMP, a novel Collaborative framework that leverages Generative Map Priors to achieve efficient compression and robust fusion. CoGMP introduces two key innovations: Element Format Feature Compression (EFFC) and Structure Guided Feature Fusion (SGFF). Specifically, EFFC leverages map element priors from codebook to encode BEV features as discrete element indices for transmitted information compression. Meanwhile, SGFF utilizes a diffusion model with structural priors to coherently integrate multi-agent features, thereby achieving consistent fusion predictions. Evaluations on the OPV2V dataset show that CoGMP achieves a 6.89/7.64 *Road/Lane IoU improvement and a* 32*-fold reduction in communication volume.*

1. Introduction

Collaborative perception aims to achieve global optimal perception with minimal message exchange between multi-agents. It has notable applications in fields such as vehicle-to-everything [54, 63] and autonomous drone swarms [45, 53]. In scenarios with multi-agents, they exchange complementary information to overcome individual constraints like restricted viewpoints [7] and occlusions [10], thereby enhancing perceptual ability. Specifically, senders need to adopt compression strategies to retain local information as much as possible under limited transmission bandwidth. Meanwhile, receivers are committed to aggregating information from senders to achieve global perception despite conflicts caused by pose errors and prediction discrepancies. To this end, collaborative perception methods focus on achieving efficient information compression on sender side and robust information fusion on receiver side.

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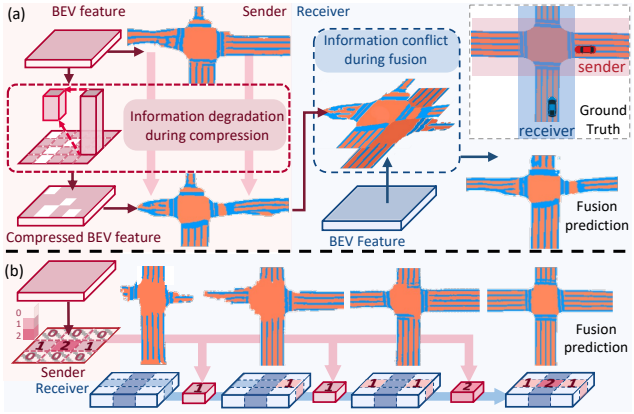


Figure 1. (a) Sender's compression suffers from information degradation, deteriorating the compressed prediction. The receiver's fusion struggles to handle information conflicts, leading to the suboptimal fusion prediction. (b) Our method leverages the generative map priors for compression in element format and fusion under structure guidance, facilitating better fusion prediction.

Existing methods [6, 17, 26, 32, 43, 49–52] primarily address the 3D object detection task in Bird's Eye View (BEV). However, accurate environment understanding requires perceiving not only surrounding objects but also scene layouts [4, 19, 23]. Therefore, we explore the compression and fusion strategies for collaborative BEV semantic segmentation, which has not been explored. Designing appropriate strategies for dense features presents significant challenges as shown in Fig. 1 (a). During the sender compresses BEV features with spatial mask and dimensionality reduction, information degradation inevitably occurs, resulting in inferior perception quality. On the receiver side, when features from multi-agents are fused, conflicting semantic estimations in overlapping regions present a substantial obstacle and lead to erroneous fusion predictions. Existing strategies tailored to collaborative detection fail to address the above challenges due to their dependency on object-level sparsity, which is absent in the dense BEV segmentation task because of non-negligible feature representation differences.

Therefore, we propose **CoGMP**, a novel **Co**llaborative framework with **G**enerative **M**ap **P**riors for BEV semantic

segmentation, aiming at efficient information compression and robust information fusion. Our method leverages the inherent *Map Priors* (*e.g.*, layout structure, road shapes, and lane continuity) from generative models to help collaborative perception. As illustrated in Fig. 1 (b), on the sender side, we introduce an innovative **E**lement **F**ormat **F**eature **C**ompression (**EFFC**) method, which utilizes element priors to represent continuous BEV features as discrete map element indices. Specifically, we first construct map element priors contained in the codebook by training the VQ-VAE [41] on BEV maps. Then, we determine the indices in the codebook by predicting the element type based on BEV features. During message transmission, replacing continuous BEV features with discrete element indices efficiently reduces communication volumes. Furthermore, senders select the critical element indices to be transmitted, which can be used by receivers to reconstruct the dropped elements. Compared with reconstructing complex high-dimensional BEV features, element priors enable easier information recovery by estimating element types. By abstracting BEV features based on element priors, EFFC significantly reduces communication volume.

On the receiver side, we propose an innovative **S**tructure **G**uided **F**eature **F**usion (**SGFF**) method, which robustly fuses features from senders under the guidance of global structural priors. In particular, SGFF progressively injects conditions from multi-agents into diffusion process [16], with the ego's preliminary perception as the initial state. Benefiting from the map structure prior learned through diffusion training, SGFF can adaptively select complementary information and resolve conflicts among agents based on implicit map structure priors. Moreover, considering the efficiency limitations of diffusion-based models when applied to perception tasks, we design an uncertainty adaptive time scheduler to enable high-quality fusion prediction with fewer iterations. By reformulating the multi-agent feature fusion as a conditional diffusion process, SGFF achieves global consistent fusion predictions.

In summary, our approach efficiently compresses information by sharing codebook element priors and leverages structure priors in the diffusion model to aggregate elements from multi-agents in appropriate regions. We validate the effectiveness of our method on the OPV2V [47] dataset, and comprehensive experiments demonstrate that CoGMP achieves state-of-the-art performance-communication trade-off. Compared to previous methods, CoGMP achieves 6.89/7.64 Road/Lane IoU improvements and a 32-fold communication volume reduction simultaneously. Additionally, improvements reach up to 10.59/11.17 Road/Lane IoU under the similar transmission bandwidth, and communication volume is further reduced by 512 factor while still maintaining superior performances. Overall, our main contributions are as follows:

- We propose a novel collaborative BEV segmentation method CoGMP that leverages generative map priors for efficient compression and robust fusion.
- We introduce EFFC for compression in the form of discrete map element indices and an SGFF process to fuse elements using a conditional diffusion model.
- We conduct extensive experiments on the OPV2V dataset, demonstrating that our method achieves a state-of-the-art performance-communication trade-off.

2. Related Works

Collaborative perception. Existing researches primarily focus on efficient information compression and robust information fusion within the collaboration process. Regarding information compression, previous methods reduce communication volume by limiting transmission targets [11, 29, 30, 57], reducing feature channels [20, 27], and using spatial filtering maps based on objectness scores [17, 18] or mutual information [50]. For robust information fusion, previous approaches employ various techniques, such as attention mechanism [26, 49, 52], graph optimization [32, 44], neural reasoning [40], and multi-scale interactions [43, 51] for multi-agent features integration. However, most methods are specifically designed for object detection task, which can not apply to BEV segmentation task. Although some studies [28, 42, 48] have developed collaborative BEV segmentation frameworks, their works typically employ task-driven compression while overlooking the utilization of structural prior information specific to maps. To fill this gap, our work introduces rich map priors from generative models, to achieve efficient information compression via element priors and robust information fusion with structure priors.

Diffusion models for perception. The recent achievements of diffusion models [16, 35] in generative tasks have sparked a growing interest in adapting for discriminative tasks. Recent works apply diffusion models to various perception tasks, including image segmentation [1, 5, 9, 21, 60], depth estimation [13, 14, 24], and object detection [8, 15, 34, 46, 56]. These studies demonstrate the effectiveness of diffusion models in addressing classical perception tasks, inspiring us to apply them to collaborative perception.

BEV segmentation. The BEV segmentation task utilizes sensor data as inputs and outputs a BEV segmentation map containing road structures such as drivable areas and lane markings. Previous approaches have mostly employed discriminative models based on CNNs [31, 37, 58] or transformers [2, 33, 61], performing per-pixel classification on BEV maps. Recently, some studies [22, 25, 59, 62, 64] have attempted to introduce generative models into BEV

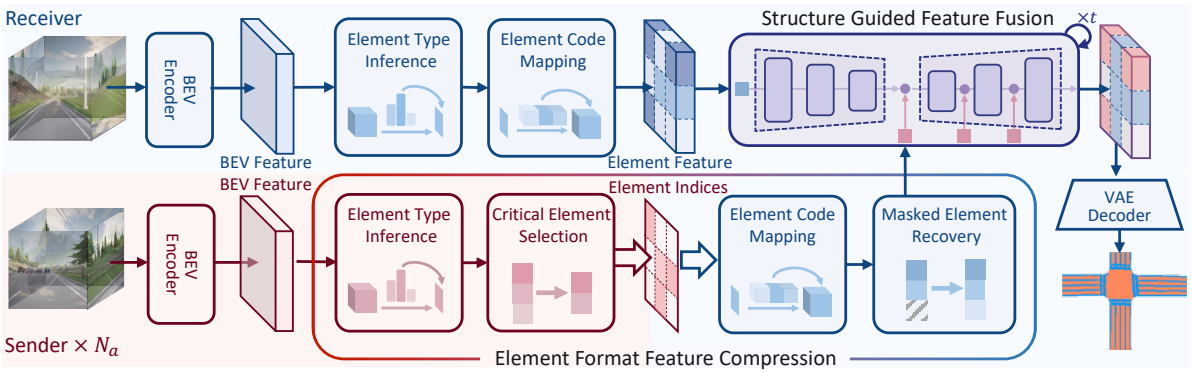


Figure 2. The overall framework of CoGMP. After extracting BEV features from surrounding multi-view images, senders perform EFFC to convert BEV features into element types and select critical elements indices as messages for transmission. These indices are used by the receiver to obtain selected element features and recover dropped element features. Next, the receiver employs SGFF to progressively refine its own element features through multiple iterations conditioned on senders' information. The fused features are finally processed by a VAE decoder to produce the collaborative BEV segmentation prediction.

segmentation. However, they are limited to the refinement of perception in a single-agent setting. In this work, we further explore the effects of generative models in multi-agent collaboration, applying them to the compression and fusion of transmitted information.

3. Method

3.1. Overview

Problem formulation. We focus on the BEV semantic segmentation task. Consider a scene with an ego agent denoted as e and N_a other agents. Let $\mathbf{X}_i$ represent the input data collected by the i-th agent's sensors (*e.g.*, multi-view cameras), and $\mathbf{Y}$ be the global ground-truth semantic label obtained from BEV maps under the ego's perspective. Our objective is to maximize the collaborative perception performance within a given communication budget B. To achieve this, the goal can be formalized as:

$$\begin{aligned} \max_{\theta,\mathcal{M}} \quad & g\left(\Phi_\theta\left(\mathbf{X}_e, \{\mathcal{M}_{i\to e}\}_{i=1}^{N_a}\right), \mathbf{Y}\right) \\ \text{s.t.} \quad & \sum_{i=1}^{N_a} b\left(\mathcal{M}_{i\to e}\right) \le B \end{aligned} \tag{1}$$

Here, $g(\cdot,\cdot)$ represents the evaluation metric, $\Phi_\theta(\cdot)$ is the collaborative perception model with trainable parameters θ, $\mathcal{M}_{i\to e}$ denotes the message transmitted from agent i to the ego agent e, and $b(\cdot)$ measures the communication volume associated with each message. The core challenge lies in determining how to select critical information to be transmitted as messages and how to utilize messages from multi-agents to enhance the ego agent's perception performance.

Framework. As shown in Fig. 2, our method utilizes a BEV encoder to generate BEV features for each agent. Then, for other agents acting as senders, we apply *Element Format Feature Compression* to use discrete key element indices as transmission units, thereby reducing communication volume. Specifically, we utilize the Element Type Inference module to compress the original continuous BEV features into discrete type indices and Element Code Mapping module to convert the indices to element features. Subsequently, the Critical Element Selection module is used to select the key elements, which enables Masked Element Recovery module to maximize the restoration of map information after the transmission. Leveraging the prior of diverse map elements, we can represent the entire map's information using a limited set of element indices. Further details can be found in Sec. 3.2.

For the ego agent acting as the receiver, we integrate the elements from other agents into the ego prediction through *Structure Guided Feature Fusion*. In particular, the receiver obtains element features from its own BEV features as mentioned above. Moreover, to aggregate element features from senders, we employ a conditional latent diffusion model which incorporates global structure priors of BEV maps. Through the diffusion reverse process, we use the features from senders as a condition to guide the progressive enhancement of the ego features. Finally, a VAE decoder is utilized to predict the segmentation result based on the fused features. Refer to Sec. 3.3 for more details.

3.2. Element Format Feature Compression

Given multi-view camera input, we use a BEV encoder following CVT [61] to extract the BEV feature map $\mathbf{F}_{bev}$ by aggregating image features via learnable queries with the assistance of intrinsic and extrinsic camera parameters.

Previous works directly transmit the continuous representation $\mathbf{F}_{bev}$ between agents, which consumes much bandwidth due to the continuous characteristic. Therefore,

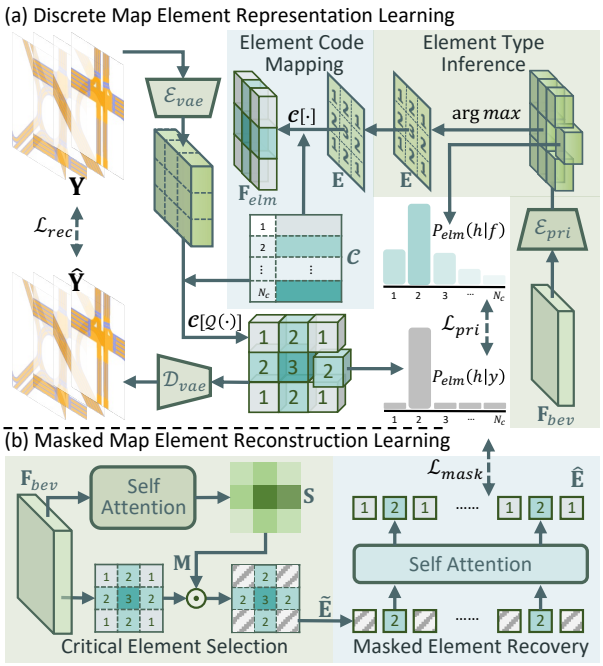


Figure 3. (a) ETI module converts BEV features into discrete element indices, enabling retrieval of map element features from the pre-built codebook via ECM module. (b) CES module selects critical elements, which are then used by the MER module to reconstruct the masked elements.

we propose **E**lement **F**ormat **F**eature **C**ompression (EFFC) to compress the continuous $\mathbf{F}_{bev}$ into masked discrete element indices $\tilde{\mathbf{E}}$ as message, which can be recovered into complete element feature $\hat{\mathbf{F}}_{elm}$ after the transmission. To achieve efficient compression based on element representation, we propose a *Discrete Map Element Representation Learning* process (seen in Fig. 3 (a)) to convert the continuous $\mathbf{F}_{bev}$ into a discrete element indices $\mathbf{E}$. Furthermore, we employ a *Masked Map Element Reconstruction Learning* process (seen in Fig. 3 (b)) to adaptively select the $\tilde{\mathbf{E}}$ of most informative elements for transmission and recover the masked elements to get $\hat{\mathbf{F}}_{elm}$.

Discrete map element representation learning. To define the map elements, we train a VQ-VAE model on the segmentation labels $\mathbf{Y}$. Similar to the original VQ-VAE, the codebook $\mathcal{C} = \{e_1, ..., e_{N_c}\}$ contains N_c vectors, where each vector is the embedding of a particular map element. The training process is almost the same as the original VQ-VAE, except that we convert the $\mathbf{Y}$ into a multi-channel binary mask image and use BCE loss as $\mathcal{L}_{rec}$ to supervise reconstruction, intending to better preserve the semantics information.

Building on the VQ-VAE model, map element priors are incorporated into the codebook vectors as element embeddings. Next, we need to learn the conversion from BEV features to element features, which can be divided into two stages: **E**lement **T**ype **I**nference (**ETI**) and **E**lement **C**ode **M**apping (**ECM**) as shown in Fig. 3 (a). For learning ETI, we first encode the $\mathbf{Y}$ to a one-hot distribution $P_{elm}(\mathbf{H}|\mathbf{Y}) \in \mathbb{R}^{h\times w\times N_c}$ as:

$$\begin{aligned} \mathcal{Q}(f) &= \arg\min_j \|f - \mathcal{C}[j]\|, \\ P_{elm}(\mathbf{H}|\mathbf{Y}) &= \text{one-hot}(\mathcal{Q}(\mathcal{E}_{vae}(\mathbf{Y}))). \end{aligned} \tag{2}$$

Then, we employed the ETI module to infer the corresponding element type indices based on BEV features. Specifically, given a BEV feature $F_{bev} \in \mathbb{R}^{h\times w\times c}$, we applied a convolutional block $\mathcal{E}_{pri}$ to obtain a element type probability distribution $P_{elm}(\mathbf{H}|\mathbf{F}_{bev}) \in \mathbb{R}^{h\times w\times N_c}$, from which an indices map $\mathbf{E}$ is generated by selecting the highest probability. This can be formulated as:

$$\begin{aligned} P_{elm}(\mathbf{H}|\mathbf{F}_{bev}) &= \mathcal{E}_{pri}(\mathbf{F}_{bev}), \\ \mathbf{E} &= \arg\max_j P_{elm}(\mathbf{H}|\mathbf{F}_{bev})[j]. \end{aligned} \tag{3}$$

Optimized by $\mathcal{L}_{pri} = \text{CE}(P_{elm}(\mathbf{H}|\mathbf{Y}), P_{elm}(\mathbf{H}|\mathbf{F}_{bev}))$, where $\text{CE}(\cdot)$ is the cross-entropy loss, we can associate the continuous BEV features with the indices of specific element type in the codebook.

Subsequently, in the ECM stage, we can directly use the indices map $\mathbf{E}$ to retrieve the corresponding embeddings from the codebook $\mathcal{C}$ to obtain the element features $\mathbf{F}_{elm} \in \mathbb{R}^{h\times w\times d}$. Utilizing the element priors contained in the codebook optimized through the VQ-VAE map reconstruction process, we achieve efficient information compression using element indices as transmission units.

Masked map element reconstruction learning. The previous compression converts the continuous BEV features to discrete element indices, significantly reducing the communication volume. However, the elements still suffer from spatial redundancy (*i.e.*, background and repetitive elements can be predicted by nearby features). Therefore, we adopt the **C**ritical **E**lement **S**election (CES) and **M**asked **E**lement **R**ecovery (MER) as Fig. 3 (b) to avoid the waste of bandwidth for redundant elements transmission.

For CES in the sender side, we apply a self-attention module over F_{bev} to obtain score map $\mathbf{S} \in \mathbb{R}^{h\times w\times 2}$ as:

$$\mathbf{S} = \text{MLP}(\text{Self-Attention}(\mathbf{F}_{bev})), \tag{4}$$

A higher value in $\mathbf{S}^{(0)}$ (*i.e.*, 0-th dimension of $\mathbf{S}$) indicates that the corresponding region contains more informative and unique features. Based on $\mathbf{S}$, we derive a selection mask $\mathbf{M} \in \{0,1\}^{h\times w}$, which identifies critical elements located in regions with high values that need to be transmitted. Specifically, the $\mathbf{M}$ can be obtained as follows:

$$\begin{cases} \mathbf{M} = \text{Gumbel-Softmax}(\mathbf{S})^{(0)} & \text{in training,} \\ \mathbf{M} = \mathbb{I}\left((i,j) \in \text{TopK}(\mathbf{S}^{(0)}, K)\right) & \text{in inference,} \end{cases} \tag{5}$$

here the indicator function $\mathbb{I}(\cdot)$ assigns $M_{i,j} = 1$ if the position (i, j) is in the top-K set, and $M_{i,j} = 0$ otherwise.

Subsequently, we determine the masked discrete element indices $\tilde{\mathbf{E}} = \mathbf{E} \odot \mathbf{M}$ for transmission. Once the receiver acquires these indices, the MER module is used to recover the masked element. MER first derive $\tilde{\mathbf{F}}_{elm}$ based on $\tilde{\mathbf{E}}$ from codebook, followed by filling the masked regions with a learnable mask token embedding $\mathbf{F}_{mask} \in \mathbb{R}^{1\times C}$. Then, a self-attention module is employed to predict the probability distribution of element indices corresponding to the masked tokens as follows:

$$\begin{aligned} \tilde{\mathbf{F}}_{elm} &= \mathbf{C}[\tilde{\mathbf{E}}] \odot \mathbf{M} + \mathbf{F}_{mask} \odot (1 - \mathbf{M}), \\ P_{elm}(\mathbf{H}|\tilde{\mathbf{F}}_{elm}) &= \mathrm{MLP}(\text{Self-Attention}(\tilde{\mathbf{F}}_{elm})), \\ \hat{\mathbf{E}} &= \tilde{\mathbf{E}} \odot \mathbf{M} + \\ &\arg\max_j P_{elm}(\mathbf{H}|\tilde{\mathbf{F}}_{elm})[j] \odot (1 - \mathbf{M}), \end{aligned} \tag{6}$$

Based on the predicted indices $\hat{\mathbf{E}}$, MER recovers masked element features by retrieving the corresponding embeddings from the codebook to get recovered features $\hat{\mathbf{F}}_{elm}$. Regarding supervision, the CES and MER modules are both optimized by the $\mathcal{L}_{mask} = \mathrm{CE}(P_{elm}(\mathbf{H}|\mathbf{Y}), P_{elm}(\mathbf{H}|\tilde{\mathbf{F}}_{elm})) \odot (1 - \mathbf{M}) + \|\mathbf{M}\|_1$, where $\|\mathbf{M}\|_1$ is used as a regularization term to prevent the CES module from degenerating into trivially selecting all elements.

By optimizing the learnable mask during the reconstruction training process, we can select the most suitable elements for transmission on the sender side. These selected critical elements possess the potential to reconstruct the global information maximally on the receiver side.

3.3. Structure Guided Feature Fusion

With messages derived from multi-agents, previous methods achieve feature fusion through various techniques (*e.g.*, GNN [44], CNN [17], transformer [48]). However, these approaches struggle to handle cross-agent conflicts arising from the limitations of individual perception. To address the above issues, we propose a novel **S**tructure **G**uided **F**eature **F**usion (SGFF) method. Our approach utilizes a diffusion model to capture the global map structure prior for generating BEV segmentation results. By progressively injecting the perceived information of multi-agents as conditions for various regions, our method produces more consistent and realistic predictions under the guidance of structural priors. In the following exposition, we first introduce the preliminaries of the diffusion model briefly, then present our *Diffusion-driven Progressive Fusion* process tailored to the collaborative BEV segmentation task.

Conditional latent diffusion model. The CLDM is designed to learn a latent space distribution $p(\boldsymbol{z}|\boldsymbol{c})$. This

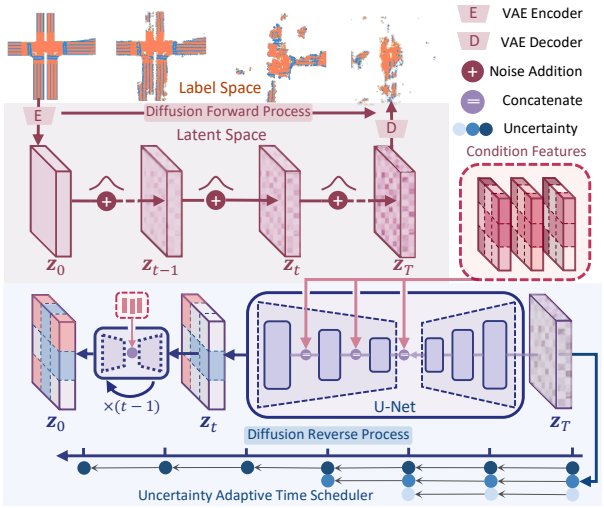


Figure 4. SGFF process pipeline. In the diffusion process, Gaussian noise is added in the forward process, and $\boldsymbol{z}_t$ is predicted by U-Net with condition features for the next reverse step. The UATS adjusts time scheduler for inference, with fewer steps required for lower uncertainty.

model assumes a forward process by gradually adding noise to the latent feature, transforming the $\boldsymbol{z}_0$ into a noisy $\boldsymbol{z}_t$ as:

$$q\left(\boldsymbol{z}_t|\boldsymbol{z}_0\right) \sim \mathcal{N}\left(\boldsymbol{z}_t; \sqrt{\bar{\alpha}_t}\boldsymbol{z}_0, \left(1-\bar{\alpha}_t\right)\mathbf{I}\right), \tag{7}$$

where $\bar{\alpha}_t = \prod_{s=1}^{t}\left(1-\beta_s\right)$, and β_s is the noise schedule.

During training, the model $f_\theta\left(\boldsymbol{z}_t, \boldsymbol{c}, t\right)$ is trained to predict the noise $\boldsymbol{\epsilon}_t$ added at each time step from $\boldsymbol{z}_t$ under the guidance of the conditional input $\boldsymbol{c}$, by minimizing the training objective function (*i.e.*, l_2 loss with the true noise).

At the inference stage, the $\boldsymbol{z}_0$ is reconstructed from random noise $\boldsymbol{z}_T$ using the model f_θ, the conditional input $\boldsymbol{c}$, and a translation rule in a Markovian manner, expressed as:

$$p_\theta\left(\boldsymbol{z}_{0:T}|\boldsymbol{c}\right) = p\left(\boldsymbol{z}_T\right)\prod_{t=1}^{T} p_\theta\left(\boldsymbol{z}_{t-1}|\boldsymbol{z}_t, \boldsymbol{c}\right). \tag{8}$$

During the reverse process, at each step t, the model f_θ predicts the noise $\boldsymbol{\epsilon}_t$ which is then used to obtain $\boldsymbol{z}_{t-1}$ by denoising $\boldsymbol{z}_t$ according to the reverse diffusion equation.

Diffusion-driven progressive fusion. Based on CLDM, we design a progressive fusion process for integrating element features from senders into the receiver features under the guidance of structure prior. As illustrated in Fig. 4, we first construct a diffusion forward process in the latent space of ground-truth semantic labels, transitioning the element features from clean $z_0 = \mathcal{E}_{vae}(\mathbf{Y})$ to noisy z_T by sequentially adding Gaussian noise as Eq. (7). This process models the gradual corruption of BEV maps, simulating the incompleteness of perception information in various regions.

Inspired by previous works [3, 38], instead of adopting ϵ-prediction for loss computation widely used in image generation, we train a lightweight U-Net [36] f_θ to directly predict the denoised feature $\boldsymbol{z}_0$ from the noisy feature $\boldsymbol{z}_t$ conditioned on $\boldsymbol{c}$ at t-th step, which can avoid the harmful variance commonly associated with $\boldsymbol{\epsilon}_t$ prediction. Here, the condition $\boldsymbol{c}$ refers specifically to the complete recovered element features $\hat{\mathbf{F}}_{elm}$ derived from senders after randomly permuting the agent order. These features are then transformed by $\mathcal{T}$ according to relative poses and processed through a convolutional block to extract the conditional features. We adopt an approach similar to ControlNet [55] to incorporate conditions, integrating them into the decoder process of U-Net. The training phase can be formalized as:

$$\begin{aligned} \boldsymbol{c} &= \mathrm{Conv}(\mathrm{Concat}(\{\mathcal{T}(\hat{\mathbf{F}}^i_{elm})\}_{i=1}^{N_a})), \\ \mathcal{L}_{ldm} &= \| \boldsymbol{z}_0 - f_\theta\left(\boldsymbol{z}_t, \boldsymbol{c}, t\right) \|^2. \end{aligned} \tag{9}$$

In the diffusion reverse process, $\boldsymbol{z}_{t-1}$ is derived from the predicted $\boldsymbol{z}_0$ following [38] for next step. To accelerate the inference, we adopt the DDIM [39] sampler to reduce the number of iteration steps. Additionally, we implement various improvements to meet the efficiency requirements of perception tasks. Rather than initializing from random noise, we start from the ego agent's element representation $\mathbf{F}^e_{elm}$, since it shares the same latent space of codebook with $\boldsymbol{z}_t$. This initialization enables inference process to utilize the ego's local perception, providing a more informed initial state. Moreover, we propose an Uncertainty Adaptive Time Scheduler (UATS) to dynamically adjust the number of iterations based on the uncertainty of ego prediction. Our design considers varying fusion complexity across scenarios (*e.g.*, multiple iterations are not necessary when ego achieves satisfactory predictions). Specifically, the uncertainty is determined by the probability distribution from ego's ETI module, and the reverse process is early-stopped when uncertainty is relatively low. This improvement significantly reduces the number of iterations required to achieve similar accuracy as shown in Tab. 3.

4. Experiments

We conduct experiments on the camera-track BEV segmentation task of the OPV2V collaborative perception dataset. We demonstrate the effectiveness and robustness of our method through quantitative evaluations in Sec. 4.2. We conduct ablation experiments to validate the impact of our proposed modules in Sec. 4.3 and performed experimental result visualization in Sec. 4.4.

4.1. Experimental Setup

Datasets. OPV2V [47] is a widely used, large-scale open simulated collaborative perception dataset. Data for OPV2V is collected from 8 towns within CARLA [12]. It comprises 73 scenes with a total of 11464 frames, with each scene lasting average 25 seconds. Each scene features 2-7 autonomous vehicles, each equipped with four directional cameras that together provide a 360-degree field of view. During testing, one autonomous vehicle is designated as the ego vehicle, with evaluations focused on a $100m \times 100m$ area centered around it and a map resolution of $39cm$.

Evaluation metric. We use Intersection-over-Union (IoU) of Road and Lane categories as our evaluation metrics, which measure segmentation accuracy by quantifying the overlap between predicted and ground-truth areas. Following previous works [18, 20, 47], communication volume is measured in terms of the number of bytes transmitted per message as $\log_2(b(\mathcal{M})/8)$.

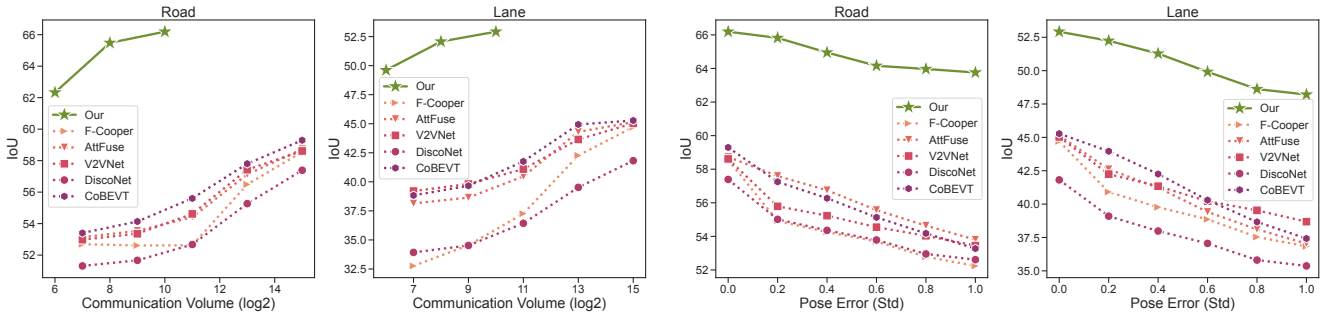
Implementation details. Following [44], we assume that all agents have a communication range of 70 meters, and agents beyond the ego's communication range do not participate in collaboration. Our training process can be divided into three stages. Firstly, we perform map-based VQ-VAE training, using latent features with a resolution of 32×32, and a codebook containing 128 element types, each with a 64-dimensional feature vector. Then, referring to the common training procedure [48], we secondly conduct single-agent training with additional $\mathcal{L}_{pri}$ and $\mathcal{L}_{mask}$ to optimize EFFC. Based on the frozen single-agent model, we thirdly train U-Net of SGFF by $\mathcal{L}_{ldm}$ to achieve multi-agent fusion. Unless otherwise specified, we use 10 iterations in SGFF.

Baseline methods. We compare with the previous state-of-the-art methods: F-Cooper [6], AttFuse [47], V2VNet [44], DiscoNet [27], and CoBEVT [48]. In order to make a fair comparison under the same communication volume cost, we adopted a BEV feature-oriented compression transmission strategy [17] for the baselines.

4.2. Quantitative Evaluation

Efficiency of EFFC compression. As shown in Fig. 5 (a), our CoGMP achieves significantly better IoU scores than previous works. To simulate various bandwidths, we conduct experiments with the communication volume ranging from 6 to 15. The experimental results demonstrate that our method achieves the best performance-bandwidth trade-off, with a SOTA performance of 66.19/52.92 Road/Lane IoU. When transmitting all intermediate features, our method improves upon the previous SOTA performance by 6.89/7.64. Under the maximum bandwidth constraint, our method achieves improvements of 8.93/10.40. Even with a bandwidth that is only $1/512$ of previous methods, our method still achieves accuracy gains of 3.03/4.33.

Robustness of SGFF fusion. We follow CoAlign [32] to add Gaussian noises $\varepsilon \in \mathcal{N}(0, \sigma)$ to the pose of each agent. The variance σ ranges from 0 to 1 and the corresponding



(a) Our CoGMP achieves the best performance-bandwidth trade-off.

(b) Our CoGMP exhibits the best performance under arbitrary pose error.

Figure 5. Comparison of the performances between our CoGMP and baselines under various communication volume and pose error.

segmentation IoU scores are plotted in Fig. 5 (b). As σ increases from 0 to 1, the road and lane IoU of our GoGMP consistently outperforms previous methods. As σ reaches 1, the IoU score of our CoGMP drops slightly to 63.76/48.21 (dropped 2.43/4.71 IoU scores), but our method still outperforms previous methods by a larger margin, with the performance gap widening to 9.96/9.52. These results suggest that our CoGMP is more robust to pose errors.

4.3. Ablation Studies

Effectiveness of SGFF. The SGFF module enhances the element feature of the ego agent by integrating elements from other agents. Without SGFF, the ego's element feature is directly decoded without fusion, resulting in Road IoU and Lane IoU scores of only 58.76 and 43.92, as shown in Tab. 1. With SGFF to fuse other agents' element features, Road IoU improves to 66.19 (+7.43), and Lane IoU rises to 52.92 (+9), demonstrating the effectiveness of SGFF. To further assess the robustness of SGFF, we vary the number of other agents (denoted as N_a). As N_a increases from 0 to 4, the IoU scores also rise, suggesting that SGFF effectively leverages additional agent information. Even without other agents ($N_a = 0$), SGFF contributes gains of 2.17 and 3.9 to Road IoU and Lane IoU, respectively. The absence of other agents changes SGFF from a conditional to an unconditional module. This result supports our claim that SGFF not only integrates external information but also uses internal map priors to refine the element feature effectively.

Number of iteration steps in SGFF. SGFF incorporates an iterative approach to refine the element feature. We examine the impact of iteration steps N_s on performance, as reported in Tab. 2. Gradually increasing N_s from 1 to 20 leads to consistent improvements in performance. Notably, Road IoU increases by 5.68 when N_s goes from 1 to 2, reaching a score of 65.28, which significantly outperforms previous methods. However, the performance gains diminish with higher values of N_s. At $N_s = 10$, doubling it to 20 yields only marginal gains of 0.13 in Road IoU and 0.02 in Lane IoU. This finding suggests that SGFF can achieve strong performance with relatively few iteration steps.

Table 1. Ablation studies on SGFF and conditions from N_a agents.

SGFF	N_a	IoU(↑)	
		Road	Lane
✗	-	58.76	43.92
✓	0	60.93	47.82
	2	65.37	52.18
	4	**66.19**	**52.92**

Table 2. Ablation studies on the iteration step N_s.

N_s	IoU(↑)	
	Road	Lane
1	59.60	45.40
2	65.28	51.82
5	65.97	52.53
10	66.13	52.75
20	**66.32**	**52.94**

Effectiveness of UATS. Although the maximum number of iteration steps in SGFF is set to 10, the results in Tab. 2 indicate that fewer steps are often sufficient to achieve high IoU scores. Based on this insight, we develop the UATS strategy, which dynamically adjusts the number of iteration steps in SGFF according to the uncertainty score in the ego-perception results. As shown in Tab. 3, this adaptive early-stopping strategy significantly reduces the average number of iteration steps by 4.76, with minimal impact on IoU scores. In fact, UATS slightly improves IoU scores by mitigating data drift, thus enhancing model stability.

Effectiveness of EFFC. For the compression module, we compare the codebook features obtained through generative training with those directly optimized for the task [20]. Under the same communication constraint of 2^8 bytes, our method achieves 65.48/52.07, outperforming 63.03/49.92 with the task-oriented codebook. The CES module selects the most critical elements by learned mask for transmission at the sender while discarding elements that are easier to recover by the receiver's MER module. Tab. 4 shows the performance improvement of our method with CES. On the sender side, we discarded 75% of the elements, leaving the remaining 25% as critical elements sent to the receiver. The

Table 3. Ablation studies on UATS strategy.

UATS	IoU(↑)		$\overline{N}_s$
	Road	Lane	
	66.13	52.75	10
✓	**66.19**	**52.92**	5.24

Table 4. Ablation studies on Selection Mask.

M	IoU(↑)	
	Road	Lane
rand.	62.77	49.58
learn.	**65.48**	**52.07**

first row in the table indicates random mask, while the second row represents our learned mask. Experimental results show that the CES module improved the performance of our method by 2.71/2.49, indicating that our selection strategy can significantly compress the transmission volume while minimizing information loss in transmission.

4.4. Qualitative Results

SGFF visualization. SGFF uses a diffusion-driven approach to progressively refine the element feature of the ego agent by incorporating both external information from other agents and internal map priors. This dual integration helps produce complete, high-fidelity predictions. As shown in the first row of Fig. 6, the initial prediction $\hat{Y}_{ego}$ displays shaky lane boundaries, while the final prediction $\hat{Y}_0$ is significantly clearer. This refinement primarily relies on SGFF's map priors, which provide essential road and lane structure knowledge. The second row of Fig. 6 illustrates SGFF's ability to enhance predictions using information from other agents. Here, the ego agent is located between two intersections, with agent a_1 positioned at the upper crossing and perceiving more details about it in $\hat{Y}_{other}$. SGFF leverages this external input to fill in missing details in $\hat{Y}_{ego}$, resulting in $\hat{Y}_0$, which successfully captures the four lanes of the upper crossing while showing only two to three lanes of the lower crossing. The third row of Fig. 6 demonstrates SGFF's capability to resolve conflicts among varying perceptions from multiple agents. In this case, the ego agent perceives the crossing as a vertical road, agent a_1 as a T-junction, and agent a_2 as a horizontal road. SGFF uses its internal map priors to synthesize these conflicting views, recognizing them as partial perceptions of a crossing. Consequently, the final prediction $\hat{Y}_0$ forms a complete crossing that closely aligns with the ground truth Y.

Visualization compared with previous methods. We show our visualization comparison in Fig. 7. Compared to other methods, the BEV map segmentation results obtained by our CoGMP demonstrate higher realism. Our model can generate clearly distinguishable map elements and coherent structures, such as smooth and uninterrupted drivable areas, as well as lane markings that are parallel, evenly spaced, and clearly distinguishable. In contrast, the perception results obtained from baselines are often ambiguous, with discontinuities or missing segments in drivable areas. Moreover, the lane markings are unclear with inconsistent lane widths, making it difficult to distinguish the lanes.

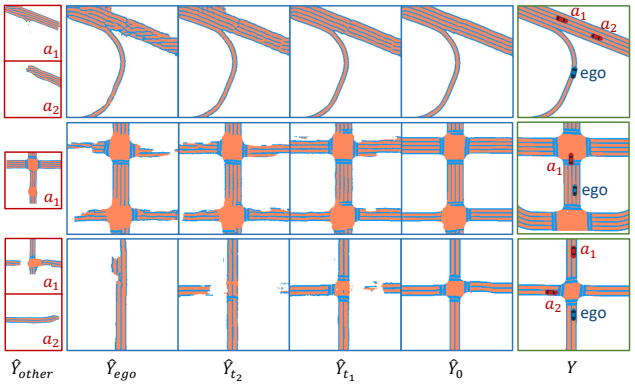


Figure 6. Visualization of SGFF. $\hat{Y}_{other}$ with a red border is the observation of other agents. Images with blue borders depict the diffusion-driven progressive fusion of SGFF, where $\hat{Y}_{ego}$ is the input, $\hat{Y}_{t_2}$ and $\hat{Y}_{t_1}$ are intermediate steps, and $\hat{Y}_0$ is the final prediction. The rightmost column labeled Y is the Ground Truth.

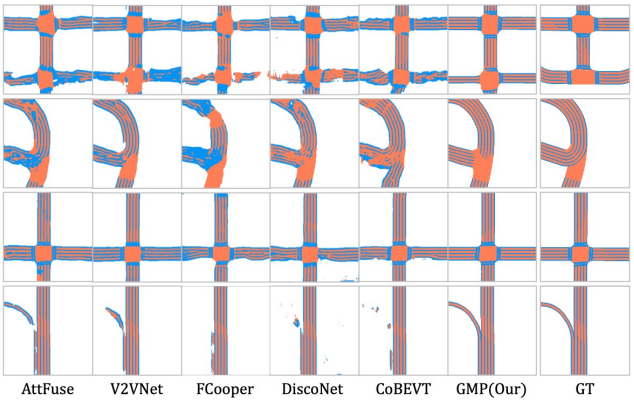


Figure 7. Qualitative comparison of BEV semantic segmentation. We show the prediction of previous methods, our CoGMP, and Ground Truth from left to right. Compared with methods without map prior, our CoGMP produce more realistic elements and reasonable structure (*e.g.*, parallel lanes and continuous roads)

5. Conclusion

This paper introduces CoGMP, a novel collaborative perception framework for BEV semantic segmentation. The core idea of the method is to leverage generative models to learn the element and structure priors of BEV maps. Benefiting from informative map priors, our method significantly reduces the communication volume during the multi-agent collaboration via efficient feature compression and achieves consistent and realistic predictions by employing a robust fusion process based on diffusion models. Our experiments show that our method achieves SOTA performance-bandwidth trade-off compared to previous methods.

6. Acknowledgement

This research is supported in part by National Key R&D Program of China (2022ZD0115502), National Natural Science Foundation of China (NO. 62461160308, U23B2010), "Pioneer" and "Leading Goose" R&D Program of Zhejiang(No. 2024C01161).

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